

Part II, corrected at the first falsifiable checkpoint

The quadratic OAR model obstructs diagonal local state-norm convergence

Lamport-form proof and source audit

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Decisive status. The Part II target in `strategy.md` is not a well-defined statement as written, because an injective wavelet map embeds *observables*; it does not push a lattice state canonically to the continuum algebra. Under the natural MMST comparison, the intended diagonal local-norm estimate is false already for the free massive scalar field.

More precisely, for every bounded interval O containing a smaller interval, there are finest-scale Weyl unitaries U_N whose lattice support and continuum wavelet support lie in O and for which

$$\liminf_{N \rightarrow \infty} |\omega_N(U_N) - \omega(\beta_N(U_N))| \geq e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} = 0.0752053795 \dots \quad (\text{R})$$

Consequently no estimate $C\varepsilon_N^\theta$, $\theta > 0$, can hold in local state norm on the diagonal scale. This does not contradict the MMST theorem: MMST take the fine-scale limit while the tested observable is held at a *fixed coarse scale*. Section 8 gives the corrected Part II target.

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1 Exact model and comparison map

This section fixes every normalization. The construction is the one-dimensional specialization of Morinelli–Morsella–Stottmeister–Tanimoto (MMST), equations (3.2)–(3.10), (4.1)–(4.5), and Proposition 4.1.

Fix $L > 0$. For $N \geq 0$, choose integers $r_N = 2^N r_0$ and put

$$\varepsilon_N = L/r_N, \quad \Lambda_N = \{\varepsilon_N n : -r_N \leq n \leq r_N - 1\} \subset \mathbb{T}_L, \quad \mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z}).$$

The dual momenta and their dimensionless representatives are

$$k_j = \frac{\pi j}{L}, \quad \theta_j = \varepsilon_N k_j = \frac{\pi j}{r_N}, \quad -r_N \leq j \leq r_N - 1. \quad (1.1)$$

For a real sequence $p = (p_n)$, define

$$\tilde{p}_j = \sum_{n=-r_N}^{r_N-1} p_n e^{-i\theta_j n}.$$

Parseval’s identity, with no suppressed constant, is

$$\sum_n |p_n|^2 = \frac{1}{2r_N} \sum_j |\tilde{p}_j|^2. \quad (1.2)$$

Let $\mathcal{W}_N$ be the Weyl algebra over $\mathbb{C}^{2r_N}$ with its standard symplectic form. In one space dimension the MMST decomposition (3.7) is $\xi = \varepsilon_N q + ip$, q, p real. Let $m > 0$ be the physical mass and put $\delta_N = m\varepsilon_N$. The nearest-neighbour lattice dispersion is

$$\gamma_N(k_j) = \left(m^2 + 2\varepsilon_N^{-2}(1 - \cos \theta_j)\right)^{1/2} \quad (1)$$

$$= \varepsilon_N^{-1} a_{\delta_N}(\theta_j), \quad a_{\delta}(\theta) = \left(\delta^2 + 4 \sin^2(\theta/2)\right)^{1/2}. \quad (1.3)$$

The centered lattice vacuum is the quasifree state

$$\omega_N(W_N(\varepsilon_N q + ip)) = \exp \left[-\frac{1}{4} \left(\|\gamma_N^{-1/2} \hat{q}\|_N^2 + \|\gamma_N^{1/2} \hat{p}\|_N^2 \right) \right], \quad (1.4)$$

where $\hat{p}_j = \varepsilon_N^{1/2} \tilde{p}_j$ and $\|z\|_N^2 = (2r_N)^{-1} \sum_j |z_j|^2$.

Let $\phi \in L^2(\mathbb{R})$ be a real compactly supported scaling function such that its integer translates are orthonormal. We assume the exact decay used in MMST,

$$|\hat{\phi}(\zeta)| \leq C(1 + |\zeta|)^{-\rho}, \quad \rho > 1, \quad (1.5)$$

which holds for the Daubechies choices admitted in MMST in one dimension. Write

$$\phi_{N,n}(x) = \varepsilon_N^{-1/2} \phi(x/\varepsilon_N - n).$$

Periodize these functions with period $2L$. The MMST symplectic map and the induced injective morphism of Weyl C^* -algebras are

$$R_N(q, p) = \varepsilon_N^{1/2} \sum_n (q_n + ip_n) \phi_{N,n}, \quad (1.6)$$

$$\beta_N(W_N(\varepsilon_N q + ip)) = W_{\text{ct}}(R_N(q, p)). \quad (1.7)$$

Let ω be the continuum mass- m vacuum on $\mathcal{W}(\mathbb{T}_L)$.

Definition 1.1 (the only canonical finite-scale comparison). The lattice state ω_N and continuum state ω are compared on $\mathcal{W}_N$ as

$$\omega_N \quad \text{versus} \quad \omega \circ \beta_N. \quad (1.8)$$

An injective $*$ -homomorphism is contravariant on states: it sends ω to $\omega \circ \beta_N$. It supplies no canonical state on $\mathcal{W}(\mathbb{T}_L)$ from ω_N .

2 Exact covariance symbols

Lemma 2.1 (lattice momentum symbol). *For every real $p \in \mathbb{R}^{2r_N}$,*

$$\omega_N(W_N(ip)) = \exp \left[-\frac{1}{4} Q_N^{\text{lat}}(p) \right], \quad Q_N^{\text{lat}}(p) = \frac{1}{2r_N} \sum_j a_{\delta_N}(\theta_j) |\tilde{p}_j|^2. \quad (2.1)$$

Proof in Lamport form

$\langle 1 \rangle 1$ Insert $q = 0$ in (1.4).

$\langle 1 \rangle 2$ Use $\hat{p}_j = \varepsilon_N^{1/2} \tilde{p}_j$ and $\gamma_N(k_j) = \varepsilon_N^{-1} a_{\delta_N}(\theta_j)$ to obtain

$$\|\gamma_N^{1/2} \hat{p}\|_N^2 = \frac{1}{2r_N} \sum_j \varepsilon_N^{-1} a_{\delta_N}(\theta_j) \varepsilon_N |\tilde{p}_j|^2,$$

which is (2.1). □

Lemma 2.2 (continuum pullback symbol). *For every real $p \in \mathbb{R}^{2r_N}$,*

$$\omega(\beta_N(W_N(ip))) = \exp \left[-\frac{1}{4} Q_N^{\text{ct}}(p) \right], \quad (2.2)$$

where

$$Q_N^{\text{ct}}(p) = \frac{1}{2r_N} \sum_j b_{\phi, \delta_N}(\theta_j) |\tilde{p}_j|^2, \quad (2.3)$$

$$b_{\phi, \delta}(\theta) = \sum_{\ell \in \mathbb{Z}} \left(\delta^2 + (\theta + 2\pi\ell)^2 \right)^{1/2} |\hat{\phi}(\theta + 2\pi\ell)|^2. \quad (2.4)$$

The series in (2.4) converges uniformly in $\theta \in [-\pi, \pi]$.

Proof in Lamport form

$\langle 1 \rangle 1$ Compute the Fourier coefficient of the embedded momentum smearing.

For continuum momentum $k \in (\pi/L)\mathbb{Z}$, (1.6) and $\widehat{\phi_{N,n}}(k) = \varepsilon_N^{1/2} e^{-ik\varepsilon_N n} \hat{\phi}(\varepsilon_N k)$ give

$$\Im \widehat{R_N(0, p)}(k) = \varepsilon_N \hat{\phi}(\varepsilon_N k) \sum_n p_n e^{-ik\varepsilon_N n}. \quad (2.5)$$

$\langle 1 \rangle 2$ Split every continuum momentum uniquely as

$$k = k_j + \frac{2\pi\ell}{\varepsilon_N}, \quad -r_N \leq j \leq r_N - 1, \quad \ell \in \mathbb{Z}. \quad (2.6)$$

Then $e^{-ik\varepsilon_N n} = e^{-i\theta_j n}$ and

$$\varepsilon_N(m^2 + k^2)^{1/2} = \left(\delta_N^2 + (\theta_j + 2\pi\ell)^2 \right)^{1/2}. \quad (2.7)$$

$\langle 1 \rangle 3$ Insert (2.5)–(2.7) into the continuum vacuum quadratic form.

The torus Fourier normalization is $(2L)^{-1} \sum_k$. Since $\varepsilon_N/(2L) = 1/(2r_N)$, grouping the sum by (2.6) gives exactly (2.3)–(2.4), with no limiting argument.

⟨1⟩ 4 Prove convergence of (2.4).

By (1.5), uniformly for $|\theta| \leq \pi$ and $|\ell| \geq 1$,

$$\left(\delta^2 + (\theta + 2\pi\ell)^2\right)^{1/2} |\widehat{\phi}(\theta + 2\pi\ell)|^2 \leq C_\delta(1 + |\ell|)^{1-2\rho}.$$

The majorant is summable because $\rho > 1$. The Weierstrass test proves uniform convergence. $\square$

Lemma 2.3 (alias partition of unity). *For every $\theta \in \mathbb{R}$,*

$$\sum_{\ell \in \mathbb{Z}} |\widehat{\phi}(\theta + 2\pi\ell)|^2 = 1. \quad (2.8)$$

Consequently, with $d(\theta) = \text{dist}(\theta, 2\pi\mathbb{Z})$,

$$b_{\phi,\delta}(\theta) \geq h_\delta(\theta) := \left(\delta^2 + d(\theta)^2\right)^{1/2}. \quad (2.9)$$

Proof in Lampion form

⟨1⟩ 1 Use orthonormality of the integer translates.

For every $n \in \mathbb{Z}$, Plancherel gives

$$\begin{aligned} \delta_{n0} &= \int_{\mathbb{R}} \phi(x) \phi(x - n) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} \left(\sum_{\ell \in \mathbb{Z}} |\widehat{\phi}(\theta + 2\pi\ell)|^2 \right) d\theta. \end{aligned} \quad (2.10)$$

The periodic function in parentheses therefore has Fourier coefficients δ_{n0} and equals 1 almost everywhere. Assumption (1.5) makes its series uniformly convergent, hence continuous. Equality almost everywhere between continuous functions implies (2.8) everywhere.

⟨1⟩ 2 For each ℓ ,

$$|\theta + 2\pi\ell| \geq d(\theta),$$

so every square-root factor in (2.4) is at least $h_\delta(\theta)$. Multiply by the nonnegative weights in (2.8) and sum. This proves (2.9). $\square$

Remark 2.4 (strict ultraviolet mismatch). For $0 < |\theta| < \pi$ and $\delta \geq 0$,

$$h_\delta(\theta) = \sqrt{\delta^2 + \theta^2} > \sqrt{\delta^2 + 4\sin^2(\theta/2)} = a_\delta(\theta), \quad (2.11)$$

because $2\sin(|\theta|/2) < |\theta|$. No property of a particular Daubechies filter can remove this inequality.

3 A localized high-frequency sequence

Fix open intervals $J \Subset O \Subset \mathbb{T}_L$ and choose real $\eta \in C_c^\infty(J)$ with $\int_{\mathbb{T}_L} \eta(x)^2 dx = 1$. Set

$$\theta_0 = \frac{3\pi}{4}, \quad p_{N,n} = c_N \sqrt{2\varepsilon_N} \eta(\varepsilon_N n) \cos(\theta_0 n), \quad (3.1)$$

where $c_N > 0$ is chosen so that $\sum_n p_{N,n}^2 = 1$. For all sufficiently large N , $2r_N$ is divisible by 8, so the carrier is periodic.

Lemma 3.1 (localized carrier measure). *For every continuous even 2π -periodic function f ,*

$$\lim_{N \rightarrow \infty} \frac{1}{2r_N} \sum_j f(\theta_j) |\tilde{p}_{N,j}|^2 = f(\theta_0). \quad (3.2)$$

Moreover $c_N \rightarrow 1$.

Proof in Lamport form

⟨1⟩ 1 Prove $c_N \rightarrow 1$.

Before multiplication by c_N^2 , the squared norm in (3.1) equals

$$2\varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \cos^2(\theta_0 n) = \varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \quad (2)$$

$$+ \varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \cos(2\theta_0 n). \quad (3.3)$$

The first sum tends to $\int \eta^2 = 1$ by Riemann integration. For the second, partition n into its four residue classes modulo 4. Since $2\theta_0 = 3\pi/2$, each residue-class Riemann sum tends to $\frac{1}{4} \int \eta^2$, while

$$\sum_{a=0}^3 e^{i(3\pi/2)a} = 0. \quad (3.4)$$

Thus the second sum tends to zero. The unnormalized squared norm tends to one, so $c_N \rightarrow 1$.

⟨1⟩ 2 Prove (3.2) first for $f(\theta) = e^{is\theta}$, $s \in \mathbb{Z}$.

By the discrete Fourier inversion formula, the left side of (3.2) is

$$\sum_n p_{N,n} p_{N,n-s}, \quad (3.5)$$

where no wrap-around term occurs for large N because $\text{supp } \eta \subseteq (-L, L)$. The product-to-sum identity turns (3.5) into

$$c_N^2 \varepsilon_N \sum_n \eta(\varepsilon_N n) \eta(\varepsilon_N(n-s)) [\cos(s\theta_0) + \cos(2n\theta_0 - s\theta_0)]. \quad (3.6)$$

The first term tends to $\cos(s\theta_0)$. For the second, split $n = 4q + a$, $a = 0, 1, 2, 3$. On each residue class the phase $e^{i(2n\theta_0 - s\theta_0)}$ equals $e^{-is\theta_0} e^{i(3\pi/2)a}$, while the corresponding Riemann sum tends to $\frac{1}{4} \int \eta(x)^2 dx$ because $\eta(x)\eta(x - \varepsilon_N s) \rightarrow \eta(x)^2$ uniformly. Summing the four limits gives zero by (3.4). Therefore (3.5) tends to

$$\cos(s\theta_0) = \frac{1}{2}(e^{is\theta_0} + e^{-is\theta_0}). \quad (3.7)$$

⟨1⟩ 3 Extend to continuous f .

Equation (3.7) proves weak convergence of the probability measures

$$\nu_N = \frac{1}{2r_N} \sum_j |\tilde{p}_{N,j}|^2 \delta_{\theta_j} \implies \frac{1}{2}(\delta_{\theta_0} + \delta_{-\theta_0}) \quad (3.8)$$

first on trigonometric polynomials. Trigonometric polynomials are uniformly dense in the continuous periodic functions, and both sides of (3.8) have mass one. Hence (3.8) holds for every continuous periodic f . If f is even, its integral against the limiting measure is $f(\theta_0)$, proving (3.2). $\square$

4 The diagonal local-norm no-go theorem

Let $X_N = \{\varepsilon_N n : p_{N,n} \neq 0\}$ and let $\mathcal{W}_N(X_N)$ be the corresponding lattice Weyl subalgebra.

Theorem 4.1 (free quadratic obstruction). *For the MMST diagonal lattice vacua and continuum mass- m vacuum,*

$$\lim_{N \rightarrow \infty} Q_N^{\text{lat}}(p_N) = 2 \sin(3\pi/8) = \sqrt{2 + \sqrt{2}}, \quad (4.1)$$

$$\liminf_{N \rightarrow \infty} Q_N^{\text{ct}}(p_N) \geq 3\pi/4. \quad (4.2)$$

Consequently

$$\liminf_{N \rightarrow \infty} \left\| \omega_N|_{\mathcal{W}_N(X_N)} - (\omega \circ \beta_N)|_{\mathcal{W}_N(X_N)} \right\| \geq c_*, \quad (4.3)$$

$$c_* = e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} > 0. \quad (4.4)$$

Numerically, $c_* = 0.0752053795 \dots$

Proof in Lamport form

$\langle 1 \rangle 1$ Apply Lemma 3.1 to the lattice symbol.

The functions a_δ converge uniformly to $a_0(\theta) = 2|\sin(\theta/2)|$, because

$$0 \leq a_\delta(\theta) - a_0(\theta) \leq \delta. \quad (4.5)$$

Equations (2.1), (3.2), and (4.5) yield

$$Q_N^{\text{lat}}(p_N) \longrightarrow a_0(\theta_0) = 2 \sin(3\pi/8) = \sqrt{2 + \sqrt{2}}. \quad (4.6)$$

$\langle 1 \rangle 2$ Apply the alias lower bound to the continuum symbol.

By (2.9),

$$Q_N^{\text{ct}}(p_N) \geq \frac{1}{2r_N} \sum_j h_{\delta_N}(\theta_j) |\tilde{p}_{N,j}|^2. \quad (4.7)$$

The periodic functions h_δ converge uniformly to $h_0(\theta) = d(\theta)$, because $0 \leq h_\delta - h_0 \leq \delta$. Lemma 3.1 and $0 < \theta_0 < \pi$ therefore give

$$\liminf_N Q_N^{\text{ct}}(p_N) \geq h_0(\theta_0) = 3\pi/4. \quad (4.8)$$

$\langle 1 \rangle 3$ Use the Weyl unitary itself as a norm-one test observable.

By (2.1)–(2.2),

$$|\omega_N(W_N(ip_N)) - \omega(\beta_N(W_N(ip_N)))| = \left| e^{-Q_N^{\text{lat}}(p_N)/4} - e^{-Q_N^{\text{ct}}(p_N)/4} \right|. \quad (4.9)$$

The right side has lower limit at least (4.4), by (4.1)–(4.2) and the strict inequality $\sqrt{2 + \sqrt{2}} < 3\pi/4$. Since $W_N(ip_N)$ is a unitary in $\mathcal{W}_N(X_N)$, the dual norm dominates (4.9). This proves (4.3). $\square$

Corollary 4.2 (no continuum extension repairs the norm). Assume $\text{supp } \phi \subset [0, R_\phi]$. For all sufficiently large N ,

$$\beta_N(\mathcal{W}_N(X_N)) \subset \mathcal{W}(O). \quad (4.10)$$

If $\tilde{\omega}_N$ is any state on the continuum Weyl algebra satisfying $\tilde{\omega}_N \circ \beta_N = \omega_N$, then

$$\liminf_{N \rightarrow \infty} \|(\tilde{\omega}_N - \omega)|_{\mathcal{W}(O)}\| \geq c_*. \quad (4.11)$$

Proof. Choose $a > 0$ with $\text{dist}(\text{supp } \eta, \mathbb{T}_L \setminus O) > a$. Every nonzero coefficient in (3.1) is based at a point of $\text{supp } \eta$, and

$$\text{supp } \phi_{N,n} \subset \varepsilon_N n + [0, R\varepsilon_N]. \quad (4.12)$$

If $R\varepsilon_N < a$, (4.10) follows. On the subalgebra in (4.10), the two functionals in (4.11) restrict to the two functionals in (4.3). Restriction cannot increase a functional norm, so (4.3) implies (4.11). $\square$

Corollary 4.3 (the proposed Part II vertical term is false). There are no constants $C < \infty$ and $\theta > 0$ for which

$$\left\| \omega_N|_{\mathcal{W}_N(X_N)} - (\omega \circ \beta_N)|_{\mathcal{W}_N(X_N)} \right\| \leq C\varepsilon_N^\theta \quad (4.13)$$

holds for all sufficiently large N .

Proof. The right side of (4.13) tends to zero; the lower limit of the left side is at least $c_* > 0$ by Theorem 4.1. $\square$

Remark 4.4 (spatial cutoffs do not evade the obstruction). In the quadratic model, let $g_R \uparrow 1$ and let the one-particle squared energies be

$$A_{N,R} = -\Delta_N + m_0^2 + \lambda g_R, \quad B_N = -\Delta_N + m_0^2 + \lambda.$$

At fixed N , $A_{N,R} \rightarrow B_N$ strongly and $m_0^2 \leq A_{N,R}, B_N \leq 4\varepsilon_N^{-2} + m_0^2 + \lambda$. Continuous functional calculus on this common compact spectral interval gives $A_{N,R}^{\pm 1/2} \rightarrow B_N^{\pm 1/2}$ strongly. Hence the expectation of every finitely supported Weyl operator converges to the full mass- m_1 lattice vacuum, $m_1^2 = m_0^2 + \lambda$. Therefore a putative estimate $\Psi(d_R) + C\varepsilon_N^\theta$, with $\Psi(d_R) \rightarrow 0$, would imply (4.13) after first taking $R \rightarrow \infty$ at fixed N . Theorem 4.1 rules this out.

5 The free facts that do survive (II.2)

Proposition 5.1 (exact gap and group-velocity bounds). *For the MMST free lattice Hamiltonian, after subtraction of the vacuum energy,*

$$\text{gap } H_N = m \quad (5.1)$$

for every N . Its one-particle group velocity satisfies

$$\sup_N \sup_{k \in [-\pi/\varepsilon_N, \pi/\varepsilon_N]} |\partial_k \gamma_N(k)| \leq 1. \quad (5.2)$$

For the quadratic spatial-cutoff model $A_N(g) = -\Delta_N + m_0^2 + \lambda g$, $0 \leq g \leq 1$, the Fock gap is at least m_0 ; on the infinite lattice with compactly supported g , it equals m_0 .

Proof in Lamport form

$\langle 1 \rangle 1$ The one-particle energy is multiplication by $\gamma_N(k)$ and $\gamma_N(k) \geq \gamma_N(0) = m$.

The normal-ordered many-body Hamiltonian is

$$H_N = d\Gamma(\gamma_N).$$

Its nonzero spectrum begins at $\inf \text{spec } \gamma_N = m$, proving (5.1).

$\langle 1 \rangle 2$ Differentiate (1.3).

$$\partial_k \gamma_N(k) = \frac{\sin(\varepsilon_N k)}{\sqrt{\delta_N^2 + 4 \sin^2(\varepsilon_N k/2)}}. \quad (5.3)$$

Since $|\sin \theta| = 2|\sin(\theta/2)\cos(\theta/2)|$, the absolute value of (5.3) is at most $|\cos(\theta/2)| \leq 1$. This proves (5.2).

$\langle 1 \rangle 3$ For a spatial cutoff, $A_N(g) \geq m_0^2$, so $A_N(g)^{1/2} \geq m_0$.

If g is compactly supported on the infinite lattice, multiplication by g is a compact perturbation of $-\Delta_N + m_0^2$. Weyl's theorem gives essential spectrum beginning at m_0^2 . The perturbation is nonnegative, so there is no spectrum below m_0^2 . Thus the bottom is exactly m_0^2 , and the Fock gap is m_0 . $\square$

Proposition 5.2 (exact Wick-constant asymptotic). *The infinite-lattice equal-site field covariance is*

$$c_N = \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{d\theta}{\sqrt{\delta_N^2 + 4\sin^2(\theta/2)}}. \quad (5.4)$$

As $N \rightarrow \infty$,

$$c_N = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1). \quad (5.5)$$

In particular the coefficient of $\log(1/\varepsilon_N)$ is exactly $1/(2\pi)$.

Proof in Lamport form

$\langle 1 \rangle 1$ Evenness and the substitution $\theta = 2u$ give

$$c_N = \frac{1}{\pi} \int_0^{\pi/2} \frac{du}{\sqrt{\delta_N^2 + 4\sin^2 u}}. \quad (5.6)$$

$\langle 1 \rangle 2$ Add and subtract the model singularity $2u$.

$$c_N = I(\delta_N) + D(\delta_N), \quad (5.7)$$

$$I(\delta) = \frac{1}{\pi} \int_0^{\pi/2} \frac{du}{\sqrt{\delta^2 + 4u^2}} = \frac{1}{2\pi} \operatorname{arsinh} \frac{\pi}{\delta}, \quad (5.8)$$

$$D(\delta) = \frac{1}{\pi} \int_0^{\pi/2} \left[\frac{1}{\sqrt{\delta^2 + 4\sin^2 u}} - \frac{1}{\sqrt{\delta^2 + 4u^2}} \right] du. \quad (5.9)$$

$\langle 1 \rangle 3$ Justify dominated convergence in (5.9).

For $0 < a \leq b$,

$$0 \leq \frac{1}{\sqrt{\delta^2 + a^2}} - \frac{1}{\sqrt{\delta^2 + b^2}} \leq \frac{1}{a} - \frac{1}{b}, \quad (5.10)$$

because the absolute derivative of $(\delta^2 + x^2)^{-1/2}$ is at most x^{-2} . Apply (5.10) with $a = 2\sin u$, $b = 2u$. The majorant

$$\frac{1}{2\sin u} - \frac{1}{2u} \quad (5.11)$$

is integrable at $u = 0$, since it equals $u/12 + O(u^3)$. Therefore

$$\lim_{\delta \downarrow 0} D(\delta) = \frac{1}{2\pi} \int_0^{\pi/2} (\csc u - u^{-1}) du \quad (3)$$

$$= \frac{1}{2\pi} \log \frac{4}{\pi}. \quad (5.12)$$

The last equality follows by integrating from a to $\pi/2$, using $\int \csc u \, du = \log \tan(u/2)$, and then taking $a \downarrow 0$.

⟨1⟩ 4 Since

$$\operatorname{arsinh}(\pi/\delta) = \log(2\pi/\delta) + o(1), \quad (5.13)$$

adding (5.8) and (5.12) yields

$$c_N = \frac{1}{2\pi} \left[\log \frac{2\pi}{\delta_N} + \log \frac{4}{\pi} \right] + o(1) = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1).$$

This proves (5.5). $\square$

Remark 5.3 (what (5.2) does and does not prove). Equation (5.2) is a group-velocity estimate. It is not, by itself, an interacting Lieb–Robinson theorem for unbounded quartic oscillators. MMST obtain a free harmonic Lieb–Robinson estimate separately in their Lemma 4.14 and Proposition 4.15. Part II must import or prove an interacting bound with the exact ε_N scaling; relabeling (5.2) as such a theorem would be invalid.

6 What the cited sources establish

Proposition 6.1 (precise MMST scope). *The following statements, and no stronger diagonal state-norm claim, are supported by the cited MMST paper.*

- (i) *The maps $\alpha_N^{N'}$ form an inductive system of observable algebras, while states at finer scales are pulled back to a fixed coarse algebra.*
- (ii) *Proposition 4.1 embeds the inductive-limit Weyl algebra into the continuum Weyl algebra by β_L .*
- (iii) *Lemmas 4.2–4.3 and Proposition 4.5 prove, for every fixed coarse scale, weak* convergence of the pulled-back free lattice ground states and identify the projective limit with the continuum vacuum on the embedded algebra.*
- (iv) *Theorem 4.11 identifies the resulting time-zero net with the massive free-field net.*
- (v) *Section 6.1 explicitly treats interacting lattice ground-state scaling limits and wavelet interacting dynamics as future generalizations.*

Proof. Items (i)–(iv) are the literal content of MMST equations (2.8)–(2.11), Propositions 4.1 and 4.5, and Theorem 4.11. Item (v) is stated in the first two paragraphs of MMST Section 6.1. Page images of all five locations appear in the separate Part II citation dossier. $\square$

Proposition 6.2 (Battle–Federbush does not supply II.4). *The 1987 Battle–Federbush note does not state $CE1_N/CE2_N$ for the MMST Daubechies OAR lattice, uniformly in N , nor for the mixed interpolation proposed in II.6.*

Proof. The note uses Meyer-type or exponentially localized orthonormal modes ψ_k and expands the Euclidean field with

$$u_k = (-\Delta + M^2)^{-1/2} \psi_k. \quad (6.1)$$

It explicitly singles out the resulting diagonality of the free action in the phase-cell variables as an attractive feature. The MMST finite-scale Hamiltonian instead has the nearest-neighbour symbol (1.3), and its Daubechies functions define symplectic scaling maps between time-zero Weyl algebras. The note contains no theorem with the MMST Hamiltonian, the Wick constant (5.4), a sharp-time GJS observable norm, an N -uniform physical gap, or two mixed vertex species. These missing hypotheses are exactly the conclusion demanded by II.4, so the note cannot be cited as that conclusion. $\square$

7 Why II.6 is not an operator interpolation

Proposition 7.1 (ill-defined mixed Hamiltonian path). *The expression*

$$H_s = (1 - s)H_N(g) + sH(g) \quad (7.1)$$

in the original II.6 program is undefined with the currently specified data.

Proof. $H_N(g)$ is a self-adjoint operator in the lattice ground-state Hilbert space $\mathcal{H}_{N,g}$, whereas $H(g)$ is a self-adjoint operator in the continuum Hilbert space $\mathcal{H}_g$. Addition of unbounded operators is defined only after both are realized as operators or closed forms on one Hilbert space with a specified common form domain. The observable embedding $\beta_N : \mathcal{W}_N \rightarrow \mathcal{W}$ supplies neither:

- (a) it maps C^* -algebra elements, not unbounded Hamiltonians;
- (b) it does not push ω_N to a canonical continuum state;
- (c) its symplectic one-particle map need not intertwine the finite-scale and continuum complex structures, so no stated second-quantized unitary identifies the two vacuum representations;
- (d) no common closed quadratic form for the two Wick-ordered interactions is given.

Therefore (7.1), its derivative $\partial_s H_s$, its spectral gap, and the vectors $D_j \Omega_s$ have no definitions. A mixed-vertex Euclidean measure is not a repair of this operator defect unless an Osterwalder–Schrader construction identifies the entire path on a common Hilbert space and proves common-domain control. $\square$

8 Corrected Part II target and dependency order

The no-go theorem forces a change of topology. It does not merely change the value of the exponent θ .

Definition 8.1 (fixed-coarse comparison). For $M < N$, a lattice state $\omega_{N,g}$ defines a state on the fixed coarse algebra $\mathcal{W}_M$ by

$$\omega_{N,g}^{[M]} := \omega_{N,g} \circ \alpha_M^N. \quad (8.1)$$

The continuum comparison state is

$$\omega_\infty^{[M]} := \omega_\infty \circ \beta_M. \quad (8.2)$$

Both sides of (8.1)–(8.2) are states on exactly the same C^* -algebra.

Definition 8.2 (resolution-separated diagonal). Let $M(N) \leq N$ satisfy

$$M(N) \rightarrow \infty, \quad N - M(N) \rightarrow \infty. \quad (8.3)$$

The first condition makes the tested observables dense in the continuum algebra; the second prevents this comparison class from containing the particular finest-scale $M = N$ carrier used in Theorem 4.1. These two conditions make the topology admissible; they do not by themselves prove convergence.

Definition 8.3 (corrected interacting Part II target). For each fixed M , bounded lattice region $X \subset \Lambda_M$, and $A \in \mathcal{W}_M(X)$,

$$\left| \omega_{N,g}^{[M]}(A) - \omega_\infty^{[M]}(A) \right| \leq \|A\| \Psi_\gamma(d) + E_{M,X,A}(N), \quad (8.4)$$

where $E_{M,X,A}(N) \rightarrow 0$ as $N \rightarrow \infty$, uniformly in the spatial cutoff once $g = 1$ on the required neighbourhood. A stronger uniform estimate may be sought along (8.3), but no supremum over all $M \leq N$ can tend to zero.

Proposition 8.4 (what is proved about the corrected target). *The left side of (8.4) is well-defined. In the free full-space case MMST prove that it tends to zero for each fixed M and A . No estimate (8.4) can hold uniformly over a class containing $M = N$ with $E_{M,X,A}(N) \rightarrow 0$ uniformly over its norm-one local observables.*

Proof. Definitions 1.1 and 8.1 put both functionals on $\mathcal{W}_M$, proving the first assertion. MMST Proposition 4.5 and Lemma 4.3 give the second assertion. For the third, take $M = N$, $X = X_N$, and the norm-one observable $A = W_N(ip_N)$ of Theorem 4.1. The left side then has lower limit at least $c_* > 0$ by (4.3), contradicting any uniform error tending to zero. $\square$

The corrected dependency order is therefore:

- II.0. Choose one lattice model.** Decide between the MMST nearest-neighbour Hamiltonian, a Daubechies Galerkin Hamiltonian, or another action. These choices have different symbols, interactions, counterterms, and locality estimates.
- II.1. Checkpoint completed.** The diagonal local-norm target fails by Theorem 4.1. Fix either the fixed-coarse topology (8.4) or a precise resolution-separated norm before proceeding.
- II.2. Free estimates partially completed.** Equations (5.1)–(5.5) give the MMST gap, group velocity, and Wick asymptotic. A separate interacting Lieb–Robinson theorem is still required.
- II.3. Source audit completed negatively.** Battle–Federbush is mechanism precedent, not $\text{CE1}_N/\text{CE2}_N$ for the chosen OAR action.
- II.4. Uniform interacting expansion remains open.** State it directly in the fixed-coarse or resolution-separated topology, including counterterms and source norms.
- II.5. LPPL with lattice tails remains open.** It must be formulated on one finite-scale algebra and must include quantitative commutator errors for affiliated unbounded Wick polynomials.
- II.6. Replace the invalid vertical interpolation.** Use a common Euclidean interpolation followed by a proved OS identification, or compare projective states directly. Do not add H_N and H across different Hilbert spaces.

9 Claim ledger

Claim	Status	Exact reason
MMST supplies the free benchmark	PASS	Fixed-coarse weak* projective limit; Proposition 4.5 and Theorem 4.11.
The displayed Part II norm is defined	FAIL	An observable embedding has no canonical push-forward on states.
Diagonal local state norm is $O(\varepsilon_N^\theta)$	FAIL	Explicit local Weyl lower bound (4.3)–(4.4).
N -uniform free gap	PASS	Exact identity gap $H_N = m$.
N -uniform free group velocity	PASS	Exact derivative bound (5.3).
Wick constant coefficient	PASS	Explicit asymptotic (5.5).
Group velocity alone gives interacting LR	FAIL	It controls only the free dispersion; an anharmonic commutator theorem is separate.
Battle supplies $\text{CE1}_N/\text{CE2}_N$	FAIL	Different variables/action and no theorem in the required norms or mixed family.

Claim	Status	Exact reason
$H_s = (1 - s)H_N + sH$ is defined	FAIL	Operators live on different Hilbert spaces; no common form realization is specified.
Fixed-coarse interacting OAR estimate	CONDITIONAL	Meaningful target; free case known pointwise, interacting uniform expansion still absent.
Full Part II OAR closure as originally written	FAIL	False topology plus undefined interpolation; it cannot be repaired by estimating a different exponent.

10 Reproducibility

The script `m2_ii_checkpoint.py` evaluates the finite-torus quadratic forms in (2.1) and the universal lower form from (2.9), the group-velocity bound (5.3), and the Wick integral (5.4). It asserts convergence to the exact constants in (4.1)–(4.4) and (5.5). The numerical output is diagnostic only; every limiting statement used in the proof was derived analytically above.

References

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- [2] G. Battle and P. Federbush, *Ondelettes and phase cell cluster expansions, a vindication*, Commun. Math. Phys. **109** (1987), 417–419, [doi:10.1007/BF01206144](https://doi.org/10.1007/BF01206144).